

Coding of Solution

Date September 2026
Team ID xxxxxx
Project Name EduMind AI - Smart AI-Powered Learning Assistant
Maximum Marks 3 Marks

Step 1: Development Approach

The solution was developed as a modular web application. The frontend collects user requests, FastAPI routes the request, feature modules prepare the AI prompt, and the Gemini service returns the generated content.

Step 2: Implemented Development Areas

S.No	Development Area	Implementation Summary	Status
1	Application Setup	FastAPI application, static files, templates and routes configured.	Implemented
2	Gemini Integration	Google Gen AI client loaded from environment variable and used for content generation.	Implemented
3	Question Answering	User questions are converted into an educational assistant prompt and sent to Gemini.	Implemented
4	Concept Explanation	Topic explanations are generated with beginner-friendly instructions.	Implemented
5	Quiz Generation	Topic-based quiz content is generated for practice.	Implemented
6	Study Notes	Structured notes are generated with key points and examples.	Implemented
7	Summarization	Provided text can be condensed into shorter study content.	Implemented
8	Learning Path	Topic and learner level are used to generate a learning sequence.	Implemented

Step 3: Configuration and Security

The Gemini API credential is stored in the local .env file and loaded through environment variables. The secret key is not hard-coded in source files or included in public project documentation.